

WEEK 6 – POWER BI REPORT

HEALTHCARE ANALYTICS DASHBOARD

Submitted By: Divya B

Roll No: E25DA013

Department: Data Science and Artificial Intelligence

Tool Used: Microsoft Power BI

Project: Healthcare Analytics – Patient Admission, Billing & Hospital
Insights

1. Introduction

The Healthcare Analytics Dashboard was developed using Microsoft Power BI to analyze patient admissions, billing information, medical conditions, insurance providers, and hospital-related trends. DAX measures were created to calculate important healthcare metrics such as total patients, total billing amount, average billing amount, average length of stay, and readmission rate. Interactive visuals and slicers were added to make the dashboard easy to analyze.

2. Problem Statement

Healthcare data contains large amounts of information related to patients, admissions, billing, medical conditions, and insurance. Analysing this information manually can be difficult and time-consuming. The objective of this project is to transform healthcare data into meaningful visual insights using Power BI.

3. Objectives

- To analyse total patient admissions.
- To calculate total healthcare billing revenue.
- To find the average billing cost per patient.
- To analyse the average length of hospital stay.
- To calculate the readmission rate.
- To analyse patients based on age group, gender, and medical condition.
- To compare billing amounts across insurance providers.
- To analyse admission and billing trends over time.
- To create an interactive healthcare dashboard using Power BI.

4. Dataset Description

The healthcare dataset contains patient, admission, medical condition, billing, insurance, and date-related information.

Important fields used:

Field	Description
Patient ID / Patient Key	Identifies the patient
Date of Admission	Patient admission date

Field	Description
Discharge Date	Patient discharge date
Age Group	Categorizes patients by age
Gender	Patient gender
Medical Condition	Patient's medical condition
Admission Type	Elective, Emergency, or Urgent
Insurance Provider	Insurance company
Billing Amount	Healthcare billing amount

5. Tools and Technologies

Tool Used: Microsoft Power BI

Technologies / Features Used:

- Power Query
- DAX
- Data Modelling
- KPI Cards
- Matrix
- Donut Chart
- Stacked Bar Chart
- Line Chart
- Clustered Column Chart
- Slicers
- Cross-filtering

6. DAX Measures

The following DAX measures were created for the dashboard.

6.1 Total Patients

Total Patients = COUNTROWS(Billing)

This measure calculates the total number of patient records/admissions in the dataset.

6.2 Total Healthcare Revenue

Total Billing Amount = SUM(Billing[Billing Amount])

This measure calculates the total billing amount from all healthcare records.

6.3 Average Billing Cost per Patient

Avg Billing Amount = AVERAGE(Billing[Billing Amount])

This measure calculates the average billing amount for the healthcare records.

6.4 Average Length of Stay

Avg Length of Stay = AVERAGEX(Billing,DATEDIFF(Billing[Date of Admission],Billing[Discharge Date],DAY))

This measure calculates the average number of days patients stayed in the hospital.

6.5 Readmission Rate

Readmission Rate =

VAR TotalCount = [Total Patients]

VAR RepeatPatients =

COUNTROWS(

```

FILTER(

    ADDCOLUMNS(

        VALUES(Billing[Patient ID]),

        "AdmissionCount",

        CALCULATE(COUNTROWS(Billing))

    ),

    [AdmissionCount] > 1

)

)

```

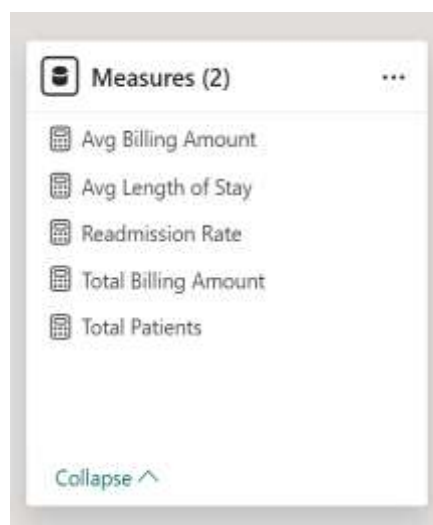
RETURN

```

DIVIDE(RepeatPatients, TotalCount, 0)

```

This measure is used to identify patients with more than one admission and calculate the readmission rate.



7. Dashboard Visualizations

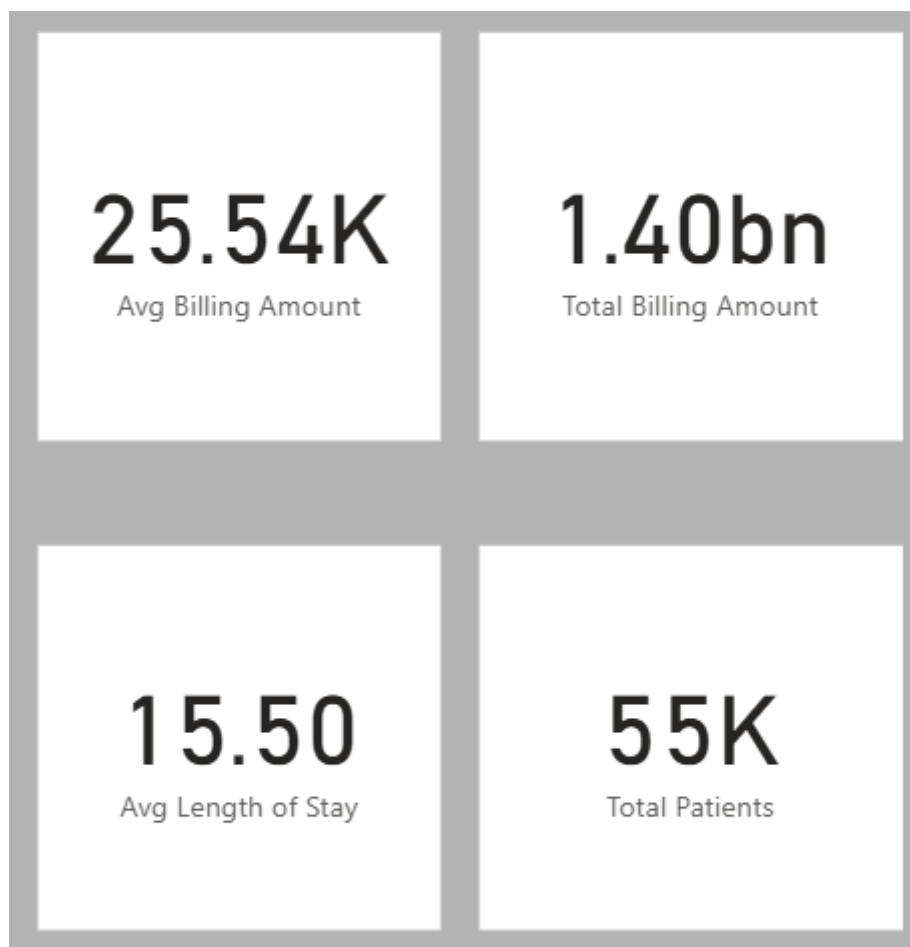
The dashboard was designed using different Power BI visuals to present healthcare information clearly.

7.1 KPI Summary Cards

Four KPI cards were created to display:

- Total Patients
- Total Billing Amount
- Average Billing Amount
- Average Length of Stay

These cards provide a quick summary of important healthcare metrics.



7.2 Patient Distribution by Age Group, Gender and Medical Condition

A matrix visual was created to analyse patient distribution based on:

- Age Group
- Gender
- Medical Condition
- Total Patients
- Average Billing Amount

This visual helps compare patient counts and billing amounts across different demographic and medical categories.

Patient Distribution by Age Group, Gender & Medical Condition						
Medical Condition	Arthritis		Asthma		Cancer	
Age Group	Total Patients	Avg Billing Amount	Total Patients	Avg Billing Amount	Total Patients	Avg Billing Am
⊕	2395	25,572.75	2477	25,561.78	2451	24,91
⊕ Elder Adult	2000	25,436.81	1933	25,387.77	1961	25,04
⊕ Senior Adult	2613	25,647.20	2570	25,697.51	2558	25,34
⊕ Teenager	212	25,449.96	189	24,979.08	186	24,93
⊕ Younger Asult	1998	25,343.22	1926	25,950.98	1984	25,31
Total	9218	25,511.78	9095	25,633.46	9140	25,15

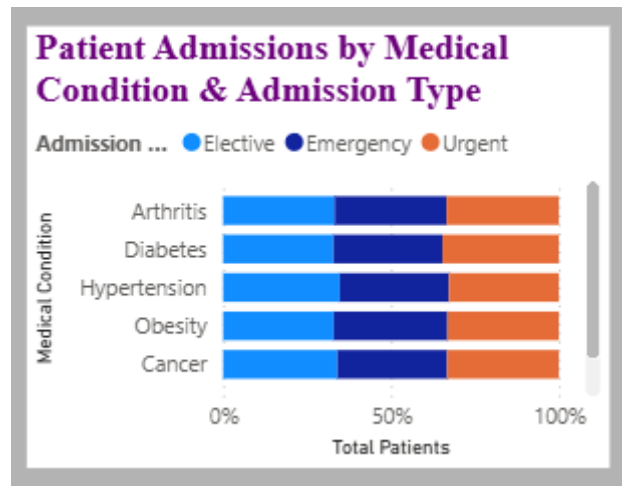
Patient Distribution by Age Group, Gender and Medical Condition

7.3 Patient Admissions by Medical Condition and Admission Type

A stacked bar chart was created using:

- **Y-Axis:** Medical Condition
- **Values:** Total Patients
- **Legend:** Admission Type

The chart compares patient admissions for different medical conditions and admission types.



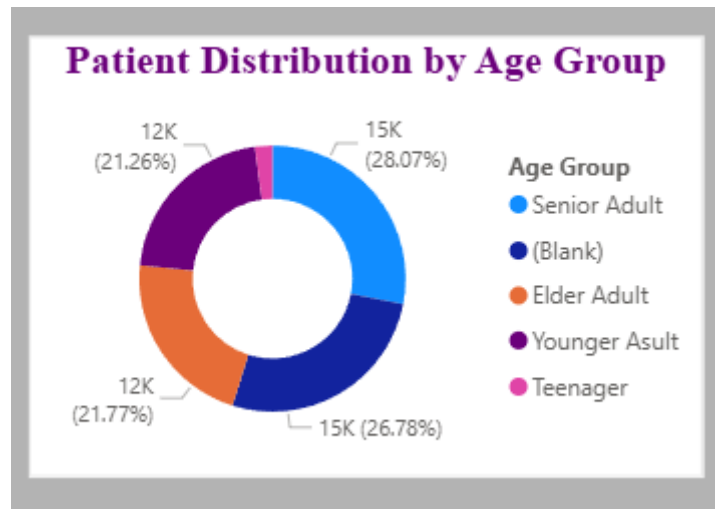
Patient Admissions by Medical Condition and Admission Type

7.4 Patient Distribution by Age Group

A donut chart was used to represent the distribution of patients across different age groups.

The age groups include:

- Teenager
- Young Adult
- Adult
- Middle Adult
- Senior Adult
- Elder Adult



Patient Distribution by Age Group

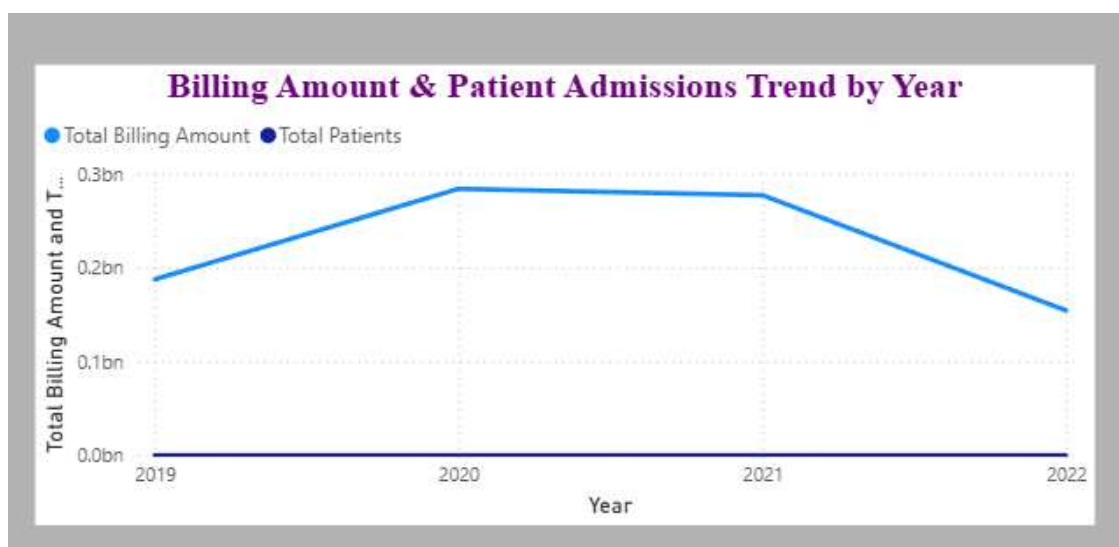
8. Admission Trends and Billing Analysis

8.1 Billing Amount and Patient Trend by Year

A line chart was created to analyse yearly healthcare trends.

- **X-Axis:** Year
- **Values:** Total Billing Amount and Total Patients

This visual helps identify changes in billing amounts and patient records over different years.



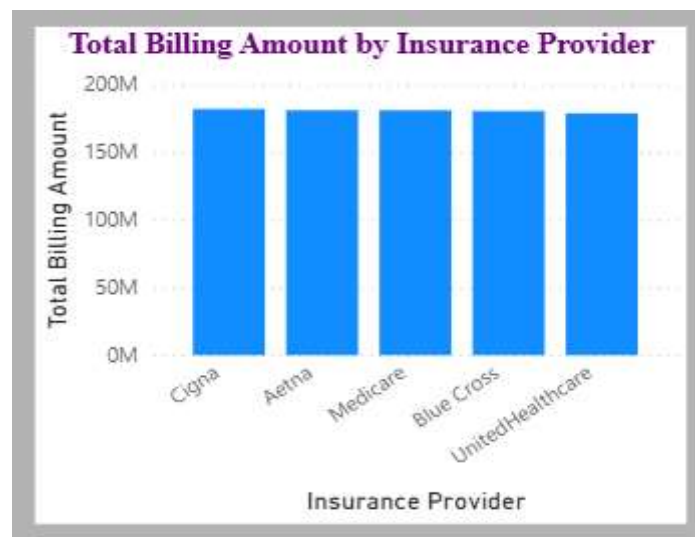
Billing Amount and Patient Admissions Trend by Year

8.2 Total Billing Amount by Insurance Provider

A clustered column chart was created to compare the total billing amount across insurance providers.

The insurance providers include:

- Cigna
- Aetna
- Medicare
- Blue Cross
- UnitedHealthcare



Total Billing Amount by Insurance Provider

9. Slicers and Dynamic Cross-Filtering

Interactive slicers were added to allow users to filter the dashboard dynamically.

Slicer 1 – Admission Date

A date range slicer was added to analyse data within a selected admission period.

The image shows a web-based slicer interface titled "Admission Date Range" in purple text. Below the title, there is a label "Date of Admission" followed by a dropdown arrow. Two date input fields are displayed: the first contains "08-05-2019" and the second contains "23-07-2022", each with a calendar icon to its right. Below these fields is a horizontal timeline slider with two circular handles, one at the start and one at the end of the range.

Admission Date Range Slicer

Slicer 2 – Insurance Provider

An insurance provider slicer was added to filter the dashboard based on the selected insurance company.

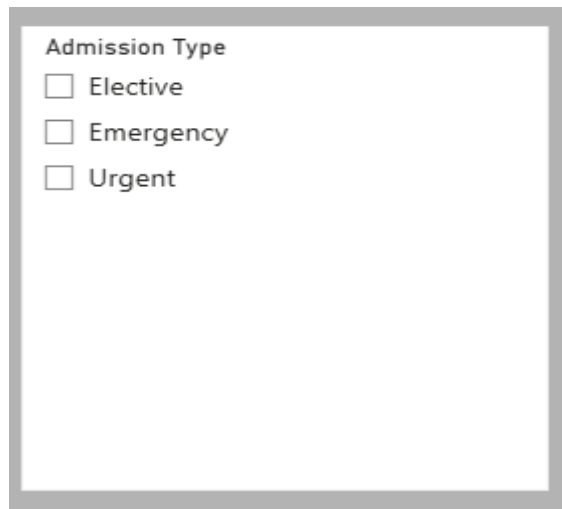
The image shows a web-based slicer interface titled "Insurance Provider" in grey text. The slicer is a grid of five white rectangular buttons with grey borders. The buttons are arranged in two rows: the top row contains "Aetna" and "Medicare", and the bottom row contains "Blue Cross" and "UnitedHealthcare". A fifth button labeled "Cigna" is positioned below "Blue Cross". The bottom-right area of the grid is a solid grey rectangle, indicating an unselected or disabled option.

Insurance Provider Slicer

Slicer 3 – Admission Type

An admission type slicer was added with:

- Elective
- Emergency
- Urgent

A rectangular box with a gray border containing a form titled "Admission Type". The form has three checkboxes, each followed by a label: "Elective", "Emergency", and "Urgent".

Admission Type

☐ Elective

☐ Emergency

☐ Urgent

Admission Type Slicer

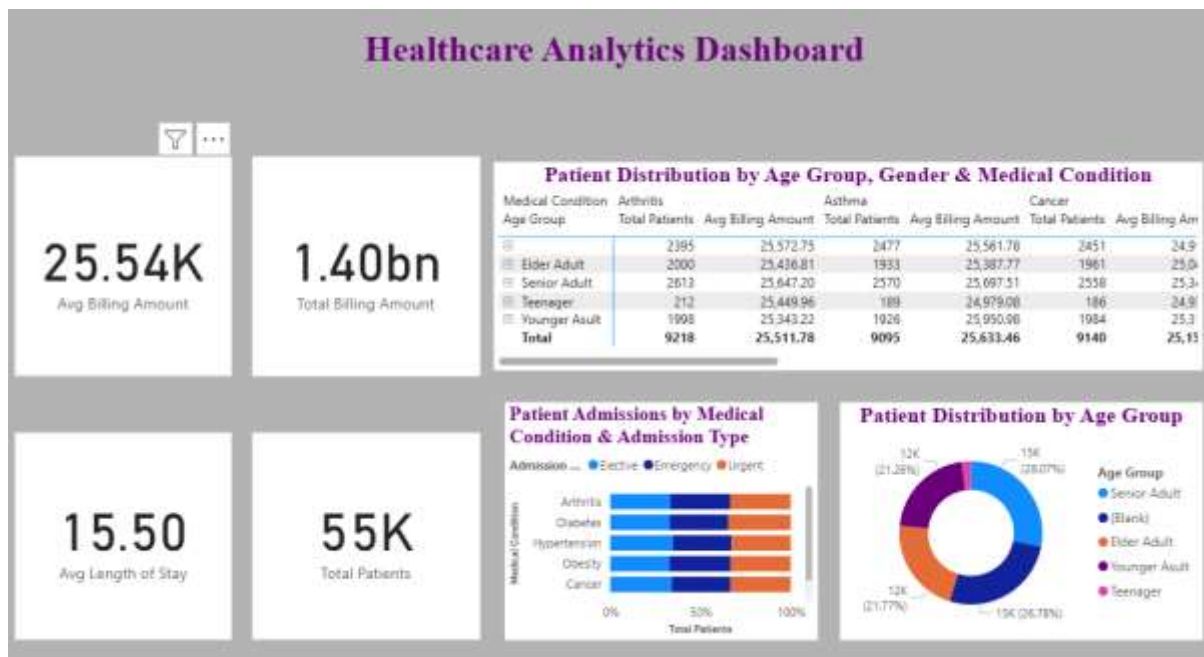
10. Dashboard – Page 1

Patient Overview & Healthcare Insights

Page 1 provides an overview of patient-related information using KPI cards, matrix, stacked bar chart, and donut chart.

It presents:

- Total patients
- Total billing amount
- Average billing amount
- Average length of stay
- Patient distribution
- Medical condition analysis
- Admission type analysis



Patient Overview and Healthcare Insights Dashboard

11. Dashboard – Page 2

Admission Trends & Billing Analysis

Page 2 focuses on time-based trends, insurance provider billing, and interactive filtering.

It contains:

- Billing and patient trend by year
- Total billing amount by insurance provider
- Admission date slicer
- Insurance provider slicer
- Admission type slicer



Admission Trends and Billing Analysis Dashboard

12. Key Insights

The dashboard provides the following insights:

- The total patient records and overall billing amount can be monitored through KPI cards.
- Average billing amount provides an overview of the typical billing value.
- Average length of stay shows the average number of days spent by patients.
- Patient distribution can be analysed across different age groups.
- Medical conditions can be compared based on patient admissions.
- Admission types such as Elective, Emergency, and Urgent can be analysed.
- Insurance providers can be compared based on total billing amount.
- Year-wise trends help understand changes in patient records and billing amounts.
- Slicers allow users to interactively filter the dashboard.

13. Conclusion

The Healthcare Analytics Dashboard successfully transforms healthcare data into an interactive and easy-to-understand Power BI report. DAX measures were used to calculate important healthcare KPIs, while different

visualizations were used to analyse patient distribution, medical conditions, admissions, billing, insurance providers, and yearly trends. The interactive slicers make it easier to explore the data based on admission date, insurance provider, and admission type.

Overall, the project demonstrates how **Microsoft Power BI, Power Query, Data Modelling, and DAX** can be used to perform healthcare data analysis and create an effective analytical dashboard.